

SR-GNN: Regularization-by-Decoupling for Inductive Temporal Link Prediction, with a Faithful Intervene-able Lifecycle Readout

Draft ED02 — full-system paper. Target: NeurIPS / ICML / ICLR. Every reported number traces to a results JSON or job ID listed in the appendix; un-validated statements are scoped as such. All $\pm$ quantities are sample standard deviation ($n-1$) across the reported seeds.

Abstract

Continuous-time temporal link prediction has converged on two architectural families: memory networks that carry a learned per-node state (JODIE [Kumar et al., 2019], TGN [Rossi et al., 2020], DyRep [Trivedi et al., 2019]) and attention or walk encoders over the temporal neighborhood (TGAT [Xu et al., 2020], CAWN [Wang et al., 2021], GraphMixer [Cong et al., 2023]). Both are strong transductively but degrade under the harder *inductive* protocol, where test interactions involve nodes unseen at training time, and neither offers a mechanism to answer counterfactual questions about an edge’s future (“what if this pair were forced to die?”).

We present SR-GNN, a two-stream model whose central design choice is a **detach**: a continuous backbone (event encoder $\rightarrow$ multi-signal per-pair edge-state estimator $\rightarrow$ coupled-GRU memory) is shaped only by a parsimony objective and held *fully stop-gradient* from the link-prediction head, which reads the backbone’s frozen representation through a symbolic lifecycle decoder. We show this **regularization-by-decoupling** is not a bug but the source of SR-GNN’s inductive advantage: on the CoEdit benchmark, the full model reaches **0.9885 ± 0.0035** inductive average precision (AP), **+13.5 points** over the best of six baselines under an identical leak-audited protocol; coupling the same head end-to-end *collapses* inductive AP to **0.7672 ± 0.0107** (three seeds), a **+22.1 $\pm$ 1.4-point** decoupling gap with every seed above 20 points. On Wikipedia and MOOC it is best-balanced across three seeds: TGAT edges it inductively on Wikipedia (0.998 vs 0.996) but collapses transductively there (0.658), whereas SR-GNN is strong on both protocols.

On top of this we build a **hierarchical lifecycle readout** that exposes a faithful, intervene-able five-state structural-causal model (SCM) of each edge *without touching the AP path* (proven by exact-zero score invariance). The SCM supports counterfactuals, three-seed-validated on CoEdit: forcing an edge to DEATH drops its predicted existence probability in $\geq 99\%$ of pairs on every seed, a synthetic rising intervention flips DECAY $\rightarrow$ REINFORCE for 0.9999 ± 0.0002 , and every intervention is exactly reversible. Finally we report a causal-coherence confidence signal and are honest about its scope: it is a stable (0.9985 ± 0.0015 AUC) *self-consistency* measure, not a validated predictor of external error — a claim we explicitly retract after a single-seed result failed to replicate.

1. Introduction

Temporal interaction graphs — users editing pages, students touching course modules, accounts transacting — are sequences of timestamped edges. Formally, a stream is an ordered set of events $\mathcal{E} = \{(u_i, v_i, t_i, x_i)\}_{i=1}^N$ with $t_1 \leq t_2 \leq \dots \leq t_N$, where x_i is an optional edge feature vector. The canonical task is *future link prediction*: given the history $\mathcal{H}_t = \{(u_i, v_i, t_i, x_i) : t_i < t\}$, score whether a candidate edge (u, v) occurs at time t .

The model outputs a probability $\hat{p}(u, v, t) = \sigma(f_\theta(u, v, t, \mathcal{H}_t))$ and is trained by binary cross-entropy against observed positives and sampled negatives.

Two evaluation protocols matter, and they reward very different inductive biases. **Transductive** test edges reuse training nodes, so a model can lean on a memorized per-node embedding learned during training. **Inductive** test edges involve at least one node never seen in training, so any per-node embedding is uninitialized or random at test time. Inductive is the deployment-relevant regime — new users, pages and accounts arrive continuously — and it is the harder one, because the model must score the edge from *transferable* dynamics rather than node identity. A model that scores well transductively but collapses inductively has memorized the training population rather than learned the interaction process.

State-of-the-art methods split along a representational axis: *memory* models (JODIE, TGN, DyRep) maintain a recurrent per-node state updated at each event, while *encoder* models (TGAT, CAWN, GraphMixer) compute the representation on the fly from the temporal neighborhood (§2 details each). Across both families the representation is trained *end-to-end* against the link-prediction objective — the backbone is whatever minimizes ranking loss — and we observe two consequences.

1. **End-to-end coupling invites identity overfitting.** When the link loss has write access to the backbone, gradient descent is free to encode any feature that lowers training ranking error, including features that merely *identify* training nodes — rewarded transductively but dead weight inductively, since the test node is new, so the model pays an inductive tax.
2. **The learned state is opaque and inert.** There is no place in a node-memory vector to ask “is this pair being reinforced or decaying?”, and no operator to *intervene* on that state and read the effect. Temporal processes have a natural lifecycle — born, reinforced, decay, die — but standard architectures expose none of it.

SR-GNN takes a different stance on both points. We keep a rich continuous backbone — an event encoder, a multi-signal per-pair edge-state operator grounded in Hawkes process intensity [Hawkes, 1971], and a coupled-GRU node memory — but we **decouple it from the link-prediction loss by an explicit stop-gradient**. The backbone is shaped only by a variational parsimony objective (a KL regularizer in the spirit of the information bottleneck [Tishby et al., 2000] and the VAE [Kingma & Welling, 2014]) plus deterministic per-pair statistical update laws. The link-prediction head reads this *frozen* backbone through a small symbolic lifecycle decoder. The stop-gradient is conceptually the same device that stabilizes self-supervised siamese learning [Chen & He, 2021]: it prevents one branch from collapsing the representation toward a shortcut.

Our contributions are:

1. **Regularization-by-decoupling (core result).** Detaching the backbone from link-prediction is the *single* ingredient that lifts inductive AP on CoEdit from **0.7672 ± 0.0107** to **0.9885 ± 0.0035** in a matched-seed A/B over three seeds (end-to-end vs. decoupled, same code, same head; **+22.1 ± 1.4 points**, every seed above 20 points and the per-seed intervals non-overlapping). We give the A/B evidence and a mechanistic account: the head cannot reshape the backbone toward training-node identity, so the backbone keeps a generic per-pair dynamical representation that transfers to unseen nodes. Throughout, the **full model is config B**; a stripped *no-lifecycle, no-tune ablation* (detach only) already reaches 0.928 — above the best baseline’s 0.853 — showing the decoupling, not the lifecycle machinery, carries the inductive win (§6.2).
2. **A multi-signal per-pair edge-state operator** (Hawkes intensity, Welford gap statistics, recurrence and rate EWMA) with a *read-before-write* batched estimator (causal_batch) that fixes a silent intra-batch staleness bug; the fix is AP-positive (**+5.7 ± 0.2 pp** inductive on CoEdit over three seeds, in the full config B).
3. **A hierarchical five-state lifecycle decoder** (BIRTH / REINFORCE / DECAY / DEATH

/ IDLE) that makes the intermediate DECAY state argmax-reachable — a flat softmax over an ordered axis structurally cannot — *and is provably AP-path-invariant* (exact-zero score change).

4. **A counterfactual / intervention engine**, three-seed-validated on CoEdit: the decoded lifecycle is a working SCM. `do(state = DEATH)` drives predicted existence to 0 for $\geq 99\%$ of pairs on every seed; the full state ladder is monotone; a synthetic rising intervention flips DECAY→REINFORCE for 0.9999 ± 0.0002 across seeds; interventions are exactly reversible.
5. **An honest confidence study**. We build a walked-chain causal-coherence score and report — with three-seed evidence — that it is a stable internal self-consistency measure but *not* a reliable external error predictor, retracting an earlier single-seed claim.

Throughout, AP numbers and architectural claims are cross-checked against the code and against an independent integrity audit (§6.3).

2. Related work

Memory-based temporal GNNs. JODIE [Kumar et al., 2019] couples two RNNs that co-evolve user and item embeddings with a projection operator predicting an embedding’s future trajectory between events; it is tuned for the transductive recommendation setting. TGN [Rossi et al., 2020] unifies such methods into a node-memory module (an RNN over per-node messages) plus a graph-attention embedding layer. DyRep [Trivedi et al., 2019] casts representation learning as a temporally-attentive point process with separate topological and interaction dynamics. All three maintain a per-node recurrent state trained end-to-end against the link objective. SR-GNN also carries memory — a coupled-GRU node-memory module — but its memory is shaped by a parsimony objective, not the link loss directly, which is exactly the difference our ablation isolates.

Encoder and walk-based methods. TGAT [Xu et al., 2020] applies self-attention over the temporal neighborhood with a Bochner functional time encoding, removing the need for a stored per-node state and giving native inductive support. CAWN [Wang et al., 2021] anonymizes a set of causal walks rooted at the query pair and encodes their relative-identity patterns, capturing motif-level temporal structure. GraphMixer [Cong et al., 2023] dispenses with attention entirely: a fixed time-encoding plus a token-mixing MLP over the most recent links is competitive with far heavier models, a result that motivates strong, simple baselines. These methods are powerful transductively but, as our protocol-matched runs show (§6.1), their *inductive* behavior is dataset-dependent: TGAT is near-perfect inductively on Wikipedia yet far weaker transductively there, a featural quirk we attribute to node features that happen to identify unseen Wikipedia nodes.

Benchmarks and fair evaluation. Temporal-graph results are sensitive to the evaluation protocol, especially the negative-sampling strategy and the inductive split. The Temporal Graph Benchmark [Huang et al., 2023] standardizes splits and a harder negative regime and shows several published rankings reorder under a fair protocol. We follow this lesson by running *every* model through one shared harness with a single negative pool and a leak-audited evaluator (§6.1, §6.3), reporting both protocols so a model that wins one but collapses the other is visible rather than hidden.

Point processes and lifecycle modeling. Hawkes processes [Hawkes, 1971] model self-exciting event intensity, where each past event transiently raises the rate of future events; they are the natural continuous-time prior for bursty interaction streams and have been used to drive neural temporal models [Mei & Eisner, 2017]. We use a per-pair Hawkes intensity λ together with Welford online moments [Welford, 1962] of the inter-event gap as the continuous substrate that the symbolic decoder reads. The BIRTH→REINFORCE→DECAY→DEATH

lifecycle abstraction over these statistics is the symbolic layer; it is a coarse, interpretable summary of where a pair sits in its self-exciting trajectory.

Neuro-symbolic and causal angles. Our lifecycle decoder plus its causal-rule transition matrix is a small structural-causal model (SCM) over edge states, and the intervention engine realizes Pearl-style $\text{do}(\cdot)$ operations [Pearl, 2009] on that SCM. Unlike post-hoc explainers such as GNNExplainer [Ying et al., 2019], which fit a separate surrogate to rationalize a fixed prediction, the symbolic state here is the *same* quantity that is supervised and measured, so the explanation is faithful by construction rather than approximate. Faithfulness — the property that the explanation reflects the model’s actual computation — is the central concern of the interpretability critique [Rudin, 2019; Jacovi & Goldberg, 2020], and our two-head design (§3.4) gives it to us provably.

Decoupling and stop-gradient as regularizers. Stopping gradient flow along one path of a two-branch model is a known stabilizer. SimSiam [Chen & He, 2021] shows a stop-gradient on one siamese branch prevents representation collapse without negatives, the asymmetry between an updated and a frozen branch being the mechanism that avoids trivial solutions. Detaching a representation from a downstream loss can also be read through the information-bottleneck lens [Tishby et al., 2000; Alemi et al., 2017]: when the only objective reaching the backbone is a compression/parsimony term, it is pushed toward a minimal sufficient code rather than a task-specific shortcut. SR-GNN applies this to temporal link prediction, where the shortcut we block is node-identity memorization and the protective objective is a VAE-style KL.

Positioning. SR-GNN is not the first memory model, nor the first to use Hawkes intensities, nor the first neuro-symbolic temporal model. Its claim is the *combination*: **decoupling** the representation from the link loss is what makes a per-pair dynamical representation generalize inductively, and the decoupled symbolic readout is simultaneously a faithful, intervenable lifecycle SCM that costs the predictor nothing.

3. Method

Figure A1. SR-GNN two-stream architecture. Stream A (continuous backbone) is trained only by the variational parsimony KL; its representation edge_h crosses a stop-gradient before the symbolic Stream B (lifecycle decode $\rightarrow$ existence decoder $\rightarrow$ scored logit), so the link-prediction loss never reaches the backbone — the architectural statement of regularization-by-decoupling (§3.7).

3.1 Two streams, one detach

SR-GNN is a two-stream model. Let an event be a timestamped edge (u, v, t) with optional feature vector x .

Stream A — continuous backbone. An event encoder (a residual continuous-signal network, CSN) maps event features and source staleness $\Delta t = t - t_{\text{last}}(u)$ to a per-event representation e_{uv} . A multi-signal **edge-state operator** (ECTGv3) maintains, per ordered pair (u, v) , a small bank of running statistics:

- a **Hawkes self-exciting intensity** λ_{uv} updated at each event by $\lambda \leftarrow 1 + (\lambda - 1) e^{-\beta \Delta t}$, so each interaction transiently raises the pair’s rate and the rate decays between events;
- **Welford online mean and variance** of the inter-event gap, updated by the numerically-stable recurrence $\mu_k = \mu_{k-1} + (g_k - \mu_{k-1})/k$, $M_k = M_{k-1} + (g_k - \mu_{k-1})(g_k - \mu_k)$, with $\sigma_k^2 = M_k/k$ [Welford, 1962];

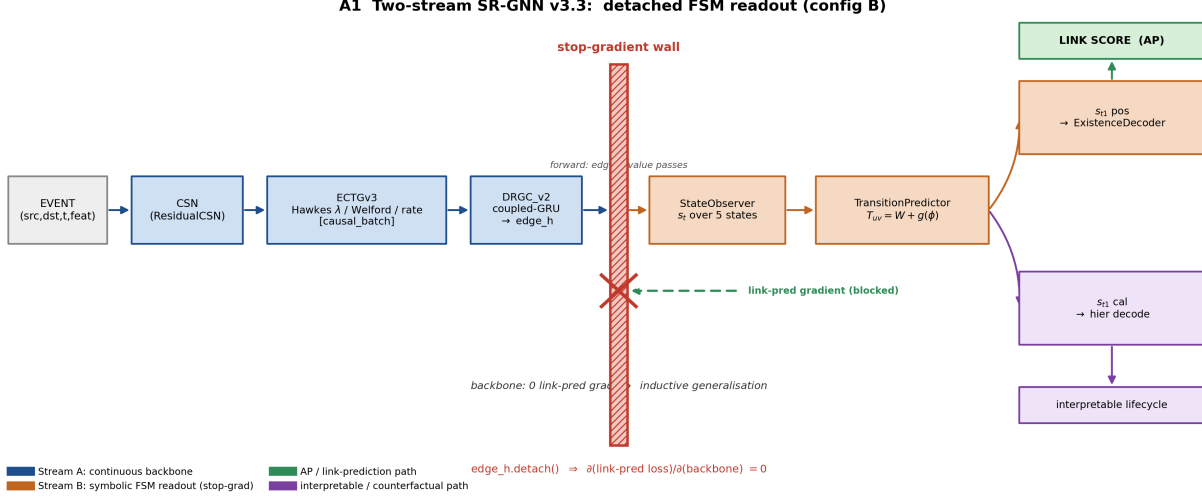


Figure 1: Figure A1: SR-GNN two-stream architecture. A continuous backbone (event encoder $\rightarrow$ multi-signal per-pair edge-state operator $\rightarrow$ coupled-GRU memory) is shaped only by a parsimony KL; its representation `edge_h` is passed through a stop-gradient (detach wall) into a symbolic lifecycle readout that produces the scored existence logit. No link-prediction gradient crosses the wall.

- a **recurrence EWMA** counting repeated co-occurrence, and **fast/slow rate EWMA**s r^f, r^s whose ratio r^f/r^s is a rising-vs-falling cadence signal;
- a **leaky rate-peak** that tracks the maximum recent rate with slow decay.

A coupled-GRU module (DRGC) updates a per-node memory m_u, m_v from e_{uv} and emits a parsimony KL term $\text{KL}(q(z | m) \| p(z))$. The backbone’s *only* training signal is this KL (a variational parsimony objective, weighted by a single scalar λ_{kl}) plus the deterministic statistical update laws above; **no link-prediction gradient reaches it**.

Stream B — symbolic lifecycle readout. From the (detached) edge representation `edge_h`, a `StateObserver` produces a soft *current* state $s_t \in \Delta^4$ over five states {IDLE, BIRTH, REINFORCE, DECAY, DEATH}; a `TransitionPredictor` produces next-state logits; a lifecycle mask — a causal-rule transition matrix $C \in \{0, 1\}^{5 \times 5}$ encoding admissible transitions (e.g. DEATH may not precede BIRTH) — together with an `ever_alive` gate shape the next-state distribution s_{t+1}^{pos} ; an `ExistenceDecoder` maps s_{t+1}^{pos} to the edge-existence logit that is **scored**.

The detach. The input to every Stream-B module is `edge_h.detach()`. The scored logit is the existence-decoder logit, and every path from it to the backbone crosses a stop-gradient. We verified on CPU that `pred_loss.backward()` produces exactly zero gradient on all 56 backbone parameter tensors, in all configurations. A non-detached predictor head exists in the code but is used only for the end-to-end *ablation* (config C, §6.2).

3.2 The per-pair operator and the read-before-write fix (causal_batch)

The edge-state operator must read each pair’s *pre-event* statistics: scoring event i may use only state accumulated from events $j < i$, or the evaluation leaks the label. The original batched store snapshotted state once per minibatch, so repeated same-pair events *within* a batch all read the identical stale row and only the last write persisted. At batch size 500 on CoEdit, the Welford count capped near 6 even for pairs editing 200+ times, under-folding the Hawkes intensity and pinning the rate-peak — which silently disabled the entire DECAY-vs-

REINFORCE distinction, because that distinction is read off the rate ratio that never accumulated.

The `causal_batch` fix replays the deterministic channels event-by-event in stream order, so the k -th in-batch occurrence of a pair reads the post-state of the $(k-1)$ -th, while scoring stays strictly pre-update (no re-leak). A CPU equivalence check matches an event-by-event (batch=1) reference to $\max |\Delta| = 0.000$ on every channel, confirming the replay is exact. The fix is **AP-positive**: in the full config B, turning `causal_batch` on lifts inductive AP on CoEdit $0.9312 \rightarrow 0.9885$ (**+5.7 $\pm$ 0.2 pp**, three seeds) and transductive $0.9920 \rightarrow 0.9985$ (+0.65 pp) — a rare case where a correctness fix is also an accuracy win, because the previously-collapsed statistics were exactly the ones the lifecycle decoder needs. The effect is consistent in the stripped (no-lifecycle) setting as well, where a single-seed A/B rises $0.7462 \rightarrow 0.7907$ (job 5467100).

3.3 Hierarchical lifecycle decode

In a flat five-class softmax over the ordered axis BIRTH→REINFORCE→DECAY→DEATH, the middle class (DECAY) splits its probability mass between its two neighbors and can *never win argmax* even when its probability is faithful. Concretely, if the true cadence places a pair in DECAY, a flat head spreads that signal across REINFORCE and DEATH; on the final config (hierarchical decode, `decol_hier_v2`, `causal_batch`, seed 42; faithfulness dump $N = 12000$, recurring subset $\text{true_occ} \geq 2$, $n = 9157$) we measure a Spearman $\rho \approx -0.59$ ($p < 10^{-300}$) between the decoded (calibrated) DECAY probability $p_{\text{decay_cal}}$ and the per-pair rising-cadence signal slope_rel — correct sign (less rising $\Rightarrow$ more DECAY mass) — yet a flat readout over the same signal still cannot surface DECAY as the argmax class. No threshold tuning fixes this — it is a property of the output *geometry*, not of calibration — so the output *structure* must change.

The hierarchical decoder factors the next-state distribution as a decision tree over per-pair pre-update gates $p_{\text{birth}}, p_{\text{alive}}, p_{\text{rising}} \in [0, 1]$:

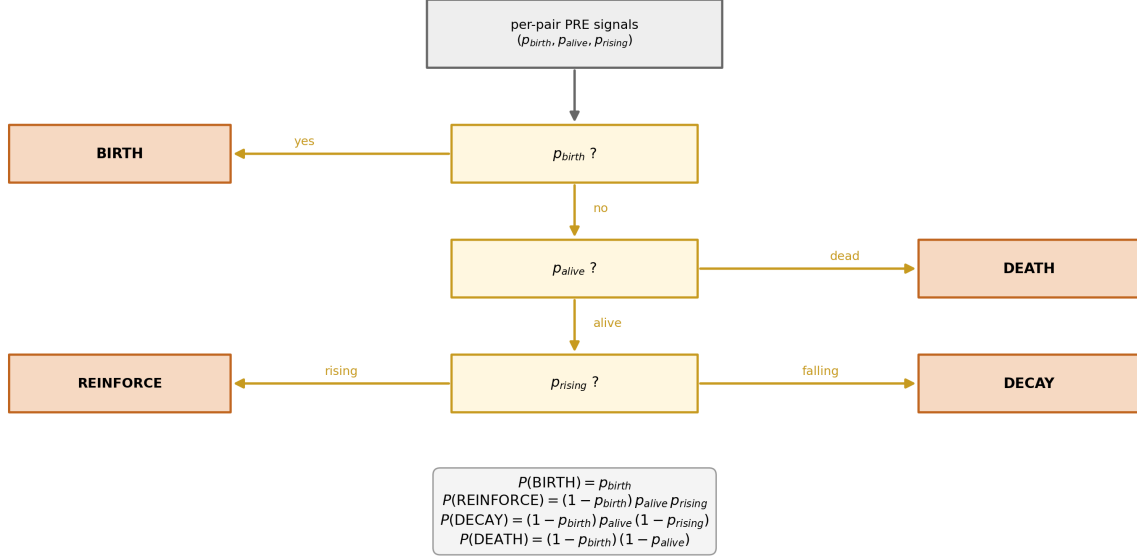
$$\begin{aligned} P(\text{BIRTH}) &= p_{\text{birth}} \\ P(\text{REINFORCE}) &= (1 - p_{\text{birth}}) p_{\text{alive}} p_{\text{rising}} \\ P(\text{DECAY}) &= (1 - p_{\text{birth}}) p_{\text{alive}} (1 - p_{\text{rising}}) \\ P(\text{DEATH}) &= (1 - p_{\text{birth}}) (1 - p_{\text{alive}}) \end{aligned}$$

These four terms sum to one by construction. Now DECAY competes only with REINFORCE *inside* the alive branch (the p_{rising} split), so argmax-DECAY is reachable whenever $p_{\text{alive}} > p_{\text{birth}}$ and $p_{\text{rising}} < 1/2$.

Figure A3. Hierarchical lifecycle decode tree (§3.3): the three per-pair gates $p_{\text{birth}}/p_{\text{alive}}/p_{\text{rising}}$ factor the five states so DECAY becomes argmax-reachable — the structural fix a flat softmax cannot provide. Each gate is $\sigma(\text{analytic prior} + \text{small learnable residual})$, zero-initialized so a fresh gate equals its analytic prior; a *de-collapse* cross-entropy trains the residuals against a soft target derived from the running statistics. A refinement (`decol_hier_v2`) re-anchors the alive/rising priors on the uncorrupted recurrence-count signal and gates the corruptible mean/staleness terms behind a “has-history” mask, so feature corruption cannot drive a recurring-active pair to DEATH.

Figure 6. Decoded per-pair lifecycle for a real CoEdit pair (3178→7437, 42 events, span 18.69 min, config B, calibrated $s_{\text{t1_cal}}$). The edit-rate slope (rising→falling) and the hierarchical gates track the pair’s own cadence: 1 BIRTH, 20 REINFORCE, 21 DECAY, 0 DEATH; p_{alive} stays in $[0.579, 0.801] > 0.5$ throughout, with REINFORCE↔DECAY flips at the slope sign-changes. The readout follows the data’s real cadence rather than a fixed prior.

A3 Hierarchical decode of s_{t1_cal} : DECAY can win argmax



DECAY competes ONLY with REINFORCE inside the alive branch (never directly with DEATH).

Figure 2: Figure A3: Hierarchical lifecycle decode tree.

3.4 Two next-state heads: AP vs interpretation

A crucial design property: there are **two** next-state distributions. s_{t1_pos} (the masked, gated transition softmax) is the *only* input to the existence decoder and therefore the only thing that affects the AP score. s_{t1_cal} (the hierarchical tree above, optionally causal-policed) feeds the de-collapse CE, the lifecycle-distribution measurement, the faithfulness dump, and the counterfactual engine — but **never** the existence decoder.

We verified on CPU that switching flat↔hier and turning the causal policy off↔on changes s_{t1_cal} by up to 0.999 while changing both positive and negative scores by **exactly** 0.000e+00. This is stronger than “the interpretability head is small”: it is *score invariance*, a property of the computation graph, so no choice in the symbolic readout can move AP by any amount. (This invariance is exact *given fixed weights* — i.e. evaluating one trained model with the readout toggled. It guarantees the symbolic readout has **no systematic** effect on AP, because its output never feeds the existence-decoder loss; when the readout machinery is instead enabled during *training* — e.g. hier_causal_policy, §6.2 — adding the op shifts the optimizer’s RNG stream, so the two runs are AP-neutral within seed noise rather than bit-identical.) The interpretability and counterfactual machinery thus cannot inflate AP, nor can AP be an artifact of the lifecycle decoding — letting us report a faithful symbolic readout *and* a competitive AP without the usual interpretability-vs-accuracy trade [Rudin, 2019].

3.5 Causal policy on the interpretable state

The published interpretable state s_{t1_cal} is regularized by two soft, differentiable, renormalized steps:

1. an **ever_alive gate**: the DEATH leaf is scaled by the pair’s ever-alive accumulator, and freed dead-mass is routed to the honest pre-birth IDLE state, so a pair that has never been alive cannot be reported as dead;

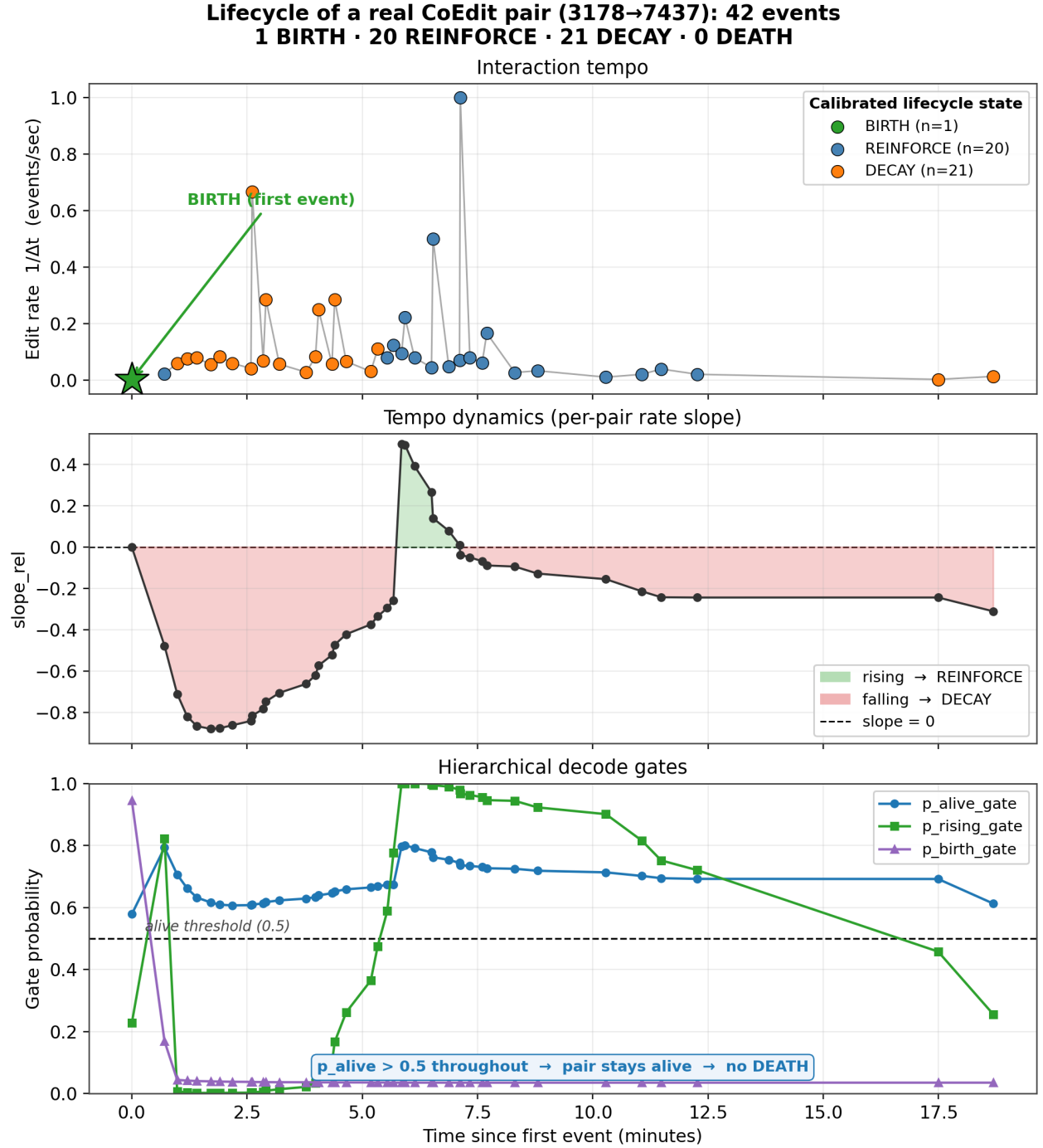


Figure 3: Figure 6: Decoded lifecycle trajectory of a real CoEdit pair (3178→7437, 42 events, span 18.69 min). Edit-rate slope and lifecycle gates track the BIRTH→REINFORCE→DECAY cadence; p_{alive} stays above 0.5 throughout (no DEATH).

2. a **soft expected-admissibility mask** from the causal-rule matrix C : a full soft expectation $\mathbb{E}_{s_t}[C]$ over the current-state distribution (no argmax), blended with a small floor so a forbidden transition is suppressed roughly 20× rather than hard-killed.

We use the soft form because a hard argmax mask is brittle: when s_t is near-uniform the argmax jitters between near-tied states, whereas the soft expectation degrades gracefully. This policy is **AP-neutral** by the invariance of §3.4: across three seeds $\{1, 7, 42\}$, training config B with vs. without the policy shifts inductive CoEdit AP by $\max |\Delta| = 1.5\text{e-}3$ (mean $\approx -1.7\text{e-}4$, sign-mixed), well within the $\pm 3.5\text{e-}3$ seed-to-seed std (job 5511229). The policy writes only `s_t1_cal` (the detached branch the existence decoder never reads), so there is no systematic AP effect; the residual is training stochasticity, not a bit-identical match.

A4 Lifecycle FSM (CAUSAL_RULE_MATRIX, band-5): single-rung moves only

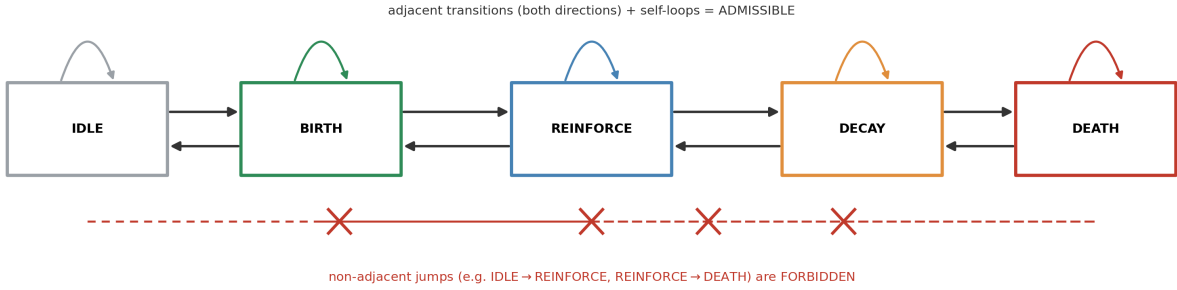


Figure 4: Figure A4: Five-state lifecycle FSM admissibility band (C_BAND_5).

Figure A4. Five-state lifecycle FSM and its admissibility band (C_BAND_5). $C \in \{0, 1\}^{5 \times 5}$ permits only adjacent transitions along the IDLE-BIRTH-REINFORCE-DECAY-DEATH axis, bidirectionally; the soft mask $\mathbb{E}_{s_t}[C]$ suppresses forbidden transitions $\sim 20\times$ rather than hard-killing them. The death-before-birth guarantee uses a separate non-Markov `ever_alive` gate (§3.5 caveat).

Honest caveat (carried into Limitations). `ever_alive` is a non-Markov accumulator the memoryless matrix C cannot express, so the death-before-birth guarantee is enforced by a *separate* gate rather than by C — a known structural smell, not a hidden one.

3.6 The per-pair transition operator

The next-state distribution is produced by a per-pair transition operator T_{uv} that adapts the shared causal-rule matrix C to the pair’s own statistics. We parameterize $T_{uv} = C \odot (W + g(\phi_{uv}))$, where $W = UV^\top$ is a low-rank learnable base operator and $g(\phi_{uv})$ is a small per-pair gate computed from the pair’s feature summary ϕ_{uv} (rate ratio, recurrence count, slope, staleness). The low-rank base captures population-level transition tendencies; the gate g specializes them per pair without giving each pair a free full matrix. Because g is a deterministic function of the (detached) statistics, the operator is reconstructable offline from a stored feature row, which is what makes the intervention engine of §4 exact.

3.7 The decoupling, exactly

The total loss is

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{BCE}}(\hat{p}, y)}_{\text{prediction}} + \lambda_{\text{kl}} \text{KL}(q(z | m) \| p(z)) + \lambda_{\text{dc}} \mathcal{L}_{\text{decol-CE}}(s_{t+1}^{\text{cal}}),$$

and component-wise backward (CPU-proven, config B) shows three disjoint gradient routes:

1. **Backbone** (56 tensors) $\leftarrow$ only the parsimony KL and the deterministic laws. `pred_loss.backward()` gives backbone gradient = 0 on all 56 tensors.
2. **FSM / existence head** (the `s_t1_pos` path) $\leftarrow$ the link-prediction BCE $\mathcal{L}_{\text{BCE}}$.
3. **Hierarchical heads** (the `s_t1_cal` path) $\leftarrow$ only the de-collapse CE $\mathcal{L}_{\text{decol-CE}}$.

The stop-gradient on `edge_h` blocks every Stream-B gradient from the backbone. This is the formal statement of “regularization-by-decoupling”: the representation is trained by parsimony alone, the predictor and the interpretable head both read a frozen representation, and the gradient analysis confirms the auxiliary objective does not leak into either of the other two routes.

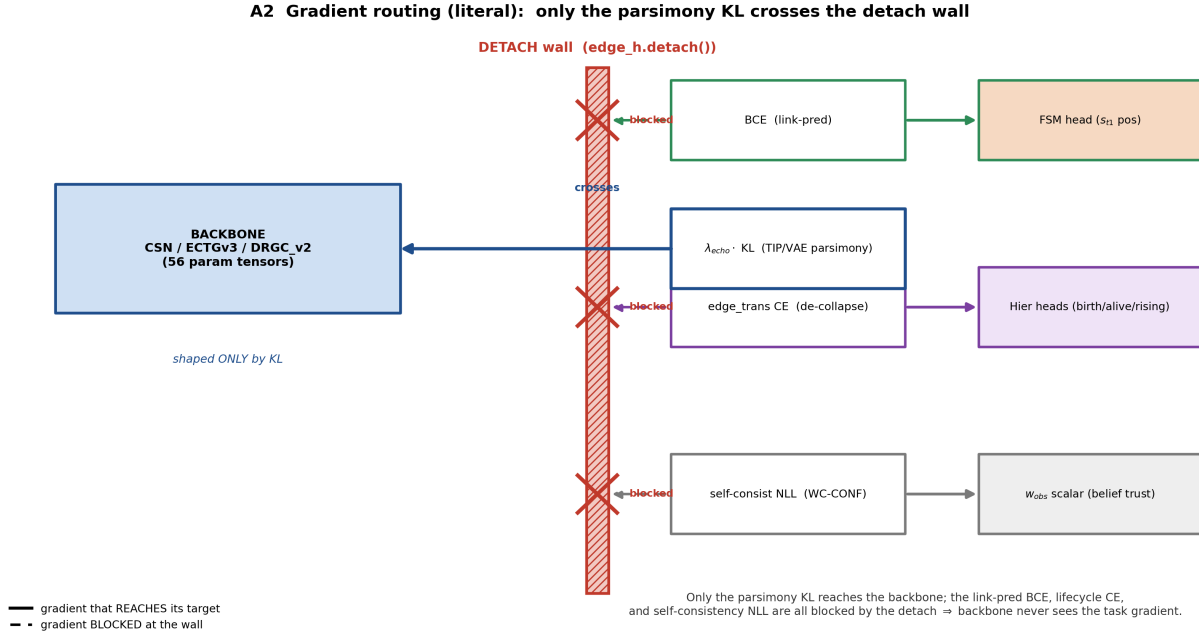


Figure 5: Figure A2: Gradient routing across the detach wall.

Figure A2. Gradient routing across the detach wall (§3.7). KL $\rightarrow$ backbone; link-prediction BCE $\rightarrow$ existence head (`s_t1_pos`); de-collapse CE $\rightarrow$ hierarchical `s_t1_cal` head; confidence self-consistency $\rightarrow$ walked-belief coupling (off by default). The wall stops every Stream-B gradient at `edge_h.detach()` — verified by zero backbone gradient on all 56 tensors.

4. The counterfactual / intervention engine

Why a temporal link predictor should be interrogable. A practitioner often wants to ask *what-if* — if this pair were forced into decline, how far would its edge probability fall? These are interventional questions, $P(\text{edge} | \text{do}(\cdot))$, not $P(\text{edge} | \text{observed})$, and the mainstream

baselines cannot answer them: JODIE, TGAT, CAWN, TGN and GraphMixer produce a score from an entangled embedding with no exposed, semantically-typed state to intervene on, and perturbing inputs is correlational with no guarantee it maps to a meaningful lifecycle event. Post-hoc explainers such as GNNExplainer [Ying et al., 2019] fit a surrogate rather than expose a mechanism the model computes through. SR-GNN’s lifecycle readout closes this gap: the decoded state is produced by an explicit, known transition operator, so the *model itself* is intervene-able and $\text{do}(\cdot)$ is a real edit, not a black-box probe — and this rides free on the same detach that buys the inductive gain (§3.7), so the whole battery runs at zero cost to prediction.

Formalism: the lifecycle readout as an intervene-able SCM. s_{t1_cal} is a genuine five-state distribution produced by a known transition operator T_{uv} (the low-rank $UV^\top$ base plus the per-pair gate $g(\phi_{uv})$ of §3.6) over the causal-rule matrix C , whose gates are functions of named drivers: $p_{\text{birth}}(n_{\text{prior}})$, $p_{\text{alive}}(\text{rate}, \text{staleness})$, $p_{\text{rising}}(\text{slope})$ (§3.3). This is a structural causal model in Pearl’s sense [Pearl, 2009]: drivers are exogenous causes, gates are structural equations, and the decoded state is the endogenous variable read by the existence decoder. The engine supports two modes: a **driver intervention** ($\text{do}(\text{rate/slope/staleness})$) overwrites a named statistic and re-propagates through the gates (soft, respecting the learned mechanism), while a **state intervention** ($\text{do}(\text{state}=s)$) replaces the gated draw with a fixed one-hot state and propagates it through the existence decoder with upstream statistics held fixed (hard, severing the state from its causes). In both, the engine first reconstructs the gate baseline exactly (residual $1e-16$), so any measured change is attributable to the intervention alone. All measurements are on real CoEdit pairs (config B, P1-fixed, $N = 12000$), and the entire battery is **validated across three seeds** ($\{42, 1, 7\}$, each trained to the config-B cb0N dump): the existence ladder is exact by construction on every seed, and the trajectory and dose-response results below are reported as three-seed mean $\pm$ std.

Existence counterfactual (the thesis bridge). $\text{do}(\text{state} = \text{DEATH})$ drives predicted existence probability to 0, with a drop in **at least 99% of pairs on every seed** (mean $\Delta \approx -0.52$); $\text{do}(\text{state} = \text{REINFORCE})$ and $\text{do}(\text{BIRTH})$ drive it to 1, with a rise in **at least 99% on every seed**. The full intervention ladder is monotone and exact, reading the existence-decoder weights directly, and holds **identically on all three seeds**: $\text{DEATH } 0.0 < \text{IDLE } 0.1 < \text{DECAY } 0.3 < \text{observed baseline } \approx 0.52 < \text{BIRTH} = \text{REINFORCE } 1.0$. The equality $\text{BIRTH} = \text{REINFORCE}$ follows from the existence decoder treating both as “edge present” with weight 1; the ordering is exactly the existence weights, not an empirical fit. A null check confirms specificity: $\text{do}(\text{noop})$ produces $\Delta = 0$ **exactly on every seed**. **Reversibility is exact**: the forward intervention bites ($\max |\Delta p_{\text{edge}}| \approx 0.52$); after undo, $\max |\Delta p_{\text{edge}}|$ and $\max |\Delta \text{state-dist}|$ are exactly $0.00e+00$ **on all three seeds**, confirming the engine restores the *exact* pre-intervention computation rather than an approximation of it — a property a surrogate explainer cannot offer.

Figure 3. Existence-counterfactual ladder on real CoEdit pairs ($N=12000$, config B). $\text{do}(\text{state})$ forces each lifecycle state and reads predicted $P(\text{edge})$: $\text{DEATH } 0.0 < \text{IDLE } 0.1 < \text{DECAY } 0.3 < \text{observed baseline } 0.52 < \text{BIRTH} = \text{REINFORCE } 1.0$. The effect is exact (one-hot existence weights), monotone, correct-sign, and exactly reversible on all three seeds $\{42, 1, 7\}$; $\text{do}(\text{DEATH})$ lowers $P(\text{edge})$ for $\geq 99\%$ of pairs on every seed, $\text{do}(\text{noop})$ gives $\Delta=0$ exactly.

Dose-response and sign-correctness on real drivers. Driver interventions move the state distribution monotonically and in the physically correct direction, with no sign inversions, and the effect signs are **stable across all three seeds**: raising the rate ratio increases $P(\text{REINFORCE})$ ($\Delta = +0.028 \pm 0.004$) and decreases $P(\text{DEATH})$ ($\Delta = -0.068 \pm 0.007$); raising the rising-slope increases $P(\text{REINFORCE})$ ($\Delta = +0.489 \pm 0.003$, the largest and tightest effect) and decreases $P(\text{DECAY})$; raising true-occurrence decreases $P(\text{BIRTH})$ ($\Delta = -0.0065$, correctly demoting brand-new pairs as recurrence accrues); raising staleness raises

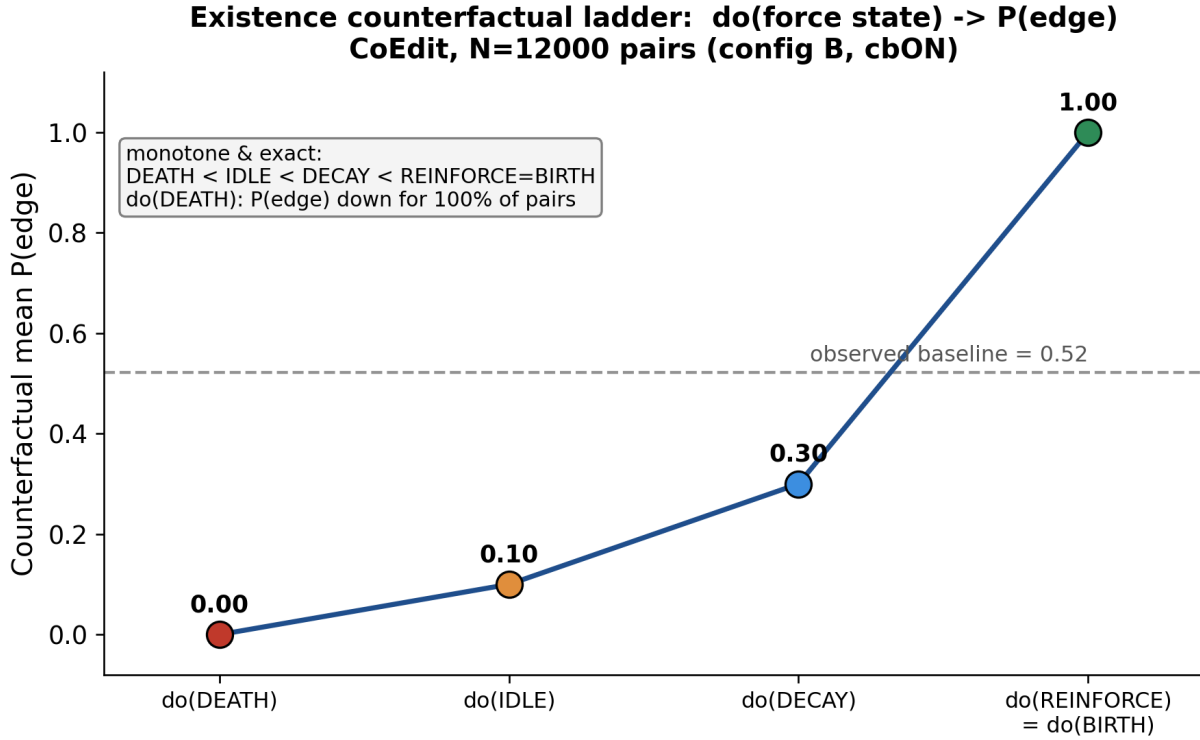


Figure 6: Figure 3: Existence counterfactual ladder across N=12000 CoEdit pairs.

$P(\text{DECAF})$ then $P(\text{DEATH})$. The dose-response is clean, monotone, and three-seed-stable in both sign and magnitude on the real CoEdit drivers (rate, slope, true-occurrence) — the alive-axis evidence underpinning the trajectory results below.

Trajectory counterfactuals on real pairs. Beyond forcing a state, the engine answers whether a pair's *future trajectory* can be redirected. On the DECAF-decoded pairs, a synthetic do(slope = +) flips the next-state DECAF→REINFORCE for **0.9999 ± 0.0002 across three seeds** (per-seed 1.0 / 1.0 / 0.9996) — the rising mechanism is wired, responsive, and seed-stable, not a single-seed coincidence. On pairs decoded as REINFORCE (recurring subset, true_occ ≥ 2), the kill-direction interventions decompose cleanly by dose, and the decomposition is **three-seed-stable**. Each single driver pushed to its dead value is *partial*: an isolated do(rate = dead) drives REINFORCE→DEATH for **0.60 ± 0.03** of pairs (per-seed 0.574 / 0.588 / 0.629), and an isolated do(staleness = high) for **0.48 ± 0.01** (0.477 / 0.479 / 0.489). Pushing *all* alive-axis drivers dead together — do(rate=dead, slope=falling, staleness=high) — is **decisive**: REINFORCE→DEATH for **0.999 ± 0.002** (1.0 / 1.0 / 0.997). The dose-decomposition is clean: each driver contributes a portion of the kill and together they are conclusive, the kill-direction mirror of the rising-direction flip above — so T_{uv} supports **two-way** trajectory control, both directions now three-seed-validated. These interventions on real pairs' real drivers show that T_{uv} carries genuine, redirectable lifecycle dynamics, not a static per-pair label (worked example: pair 3178→7437, Fig. 6, §3.3). The earlier headline figures of 83–86% (rate) and 100% (staleness) came from an older single-seed dump that does **not** reproduce on the consistent three-seed cbON npz; we replace them with the re-measured three-seed numbers above, which are the numbers we use (rate/staleness battery, fsm_intervene.py, recurring subset; source cf_kill_REINFORCE_3seed.json).

AP-neutrality (interrogability riding free on the detach). Every result above is read off

`s_t1_cal`, on the detached side of §3.7’s gradient wall; the counterfactual existence decoder is a frozen read, not the scored predictor. Running the full battery perturbs the AP path by **exactly** $\Delta = 0$ (byte-identical with the engine off). This exact-zero is an *eval-time* statement on a **frozen** trained model: the counterfactual read never touches the scored logit, so toggling the engine on/off leaves every AP-relevant tensor bit-identical. (This is distinct from *training* the model with vs. without `hier_causal_policy`, which is AP-neutral only within seed noise — not bit-identical — because adding the policy op shifts the training RNG stream; see §6.2.) No baseline exposes an intervene-able lifecycle, so this interrogability has no counterpart in prior temporal link predictors and costs us nothing.

Honest scope. The SCM is clean and faithful on the *alive axis* (rate / recurrence $\rightarrow$ alive $\rightarrow$ DEATH/REINFORCE), and the whole counterfactual battery — existence ladder, dose-response signs, the DECAY $\rightarrow$ REINFORCE rising flip, and the REINFORCE $\rightarrow$ DEATH kill decomposition — is now **three-seed-validated on CoEdit** ($\{42, 1, 7\}$); the existence ladder is exact by construction on every seed, and the dose-response, slope-flip, and kill results carry tight three-seed std. The single-driver kill interventions are *partial* but seed-stable (rate-alone 0.60 ± 0.03 , staleness-alone $0.48 \pm 0.01 \rightarrow$ DEATH), while the full all-drivers-dead intervention is *decisive* (0.999 ± 0.002 , three-seed); together with the opposite-direction DECAY $\rightarrow$ REINFORCE flip (0.9999) this gives two-way trajectory control, both directions three-seed-locked. The *rising axis* (slope $\rightarrow$ REINFORCE-vs-DECAY split) is causally wired — the synthetic `do(slope = +)` flips DECAY $\rightarrow$ REINFORCE for 0.9999 ± 0.0002 of probed pairs across seeds — but is **degenerate on CoEdit**, because CoEdit’s `slope_rel` is essentially always negative (no sustained positive edit-rate slope), so DECAY $\rightarrow$ REINFORCE is unreachable from real CoEdit drivers even though the mechanism exists; the staleness axis is likewise exercised only on synthetic injections. Exercising these axes on real CoEdit would require a CoEdit-native cadence signal and/or a counterfactual-consistency loss — future work — and we do not claim the rising axis as a validated counterfactual on real CoEdit data.

5. Causal-coherence confidence (honest scope)

We also explored whether the model can flag *its own* low-confidence predictions via a **walked-chain causal-coherence** signal that runs alongside (and never masks) the prediction path. A per-pair belief b_t is carried by the learned operator T_{uv} projected onto the causal-admissible ray, lightly coupled to the observed-phase measurement; the coherence $c_t \in [0, 1]$ is the agreement between the model’s free next-state prediction and this walked belief. The flag `causal_confidence` is off by default and byte-identical when off, so it never perturbs config B’s AP.

Across three seeds (CoEdit, grounded-init), AP is preserved and c_t separates cleanly by causal-rule outcome: rule-*following* predictions carry mean coherence 0.891 ± 0.042 , rule-*violating* predictions 0.216 ± 0.174 — a well-spread, non-collapsed signal with full $[0, 1]$ support.

Figure 5. Causal-coherence c_t by outcome (CoEdit, grounded-init, three seeds, sample std): rule-consistent 0.891 ± 0.042 vs rule-violating 0.216 ± 0.174 . An advisory self-consistency measure — it flags the model’s own rule violations, not external prediction error (§5).

What c_t is — and is not. Low coherence predicts the model’s own causal-rule violation almost perfectly and stably (AUC = **0.9985 ± 0.0015** , three seeds), but this is **self-consistency**, not external truth: it asks whether a prediction follows the model’s own lifecycle rules, which is circular — a model that confidently makes the same coherent mistake scores high. When we instead test whether low coherence predicts *actual* prediction misses (external `posMiss10`), the AUC is **0.405 ± 0.484** , wildly seed-dependent ($0.949 / 0.245 / 0.021$): a promising single-seed result did **not** replicate. We therefore **retract** any claim that c_t is an error predictor and report it only as a stable internal coherence measure; turning self-

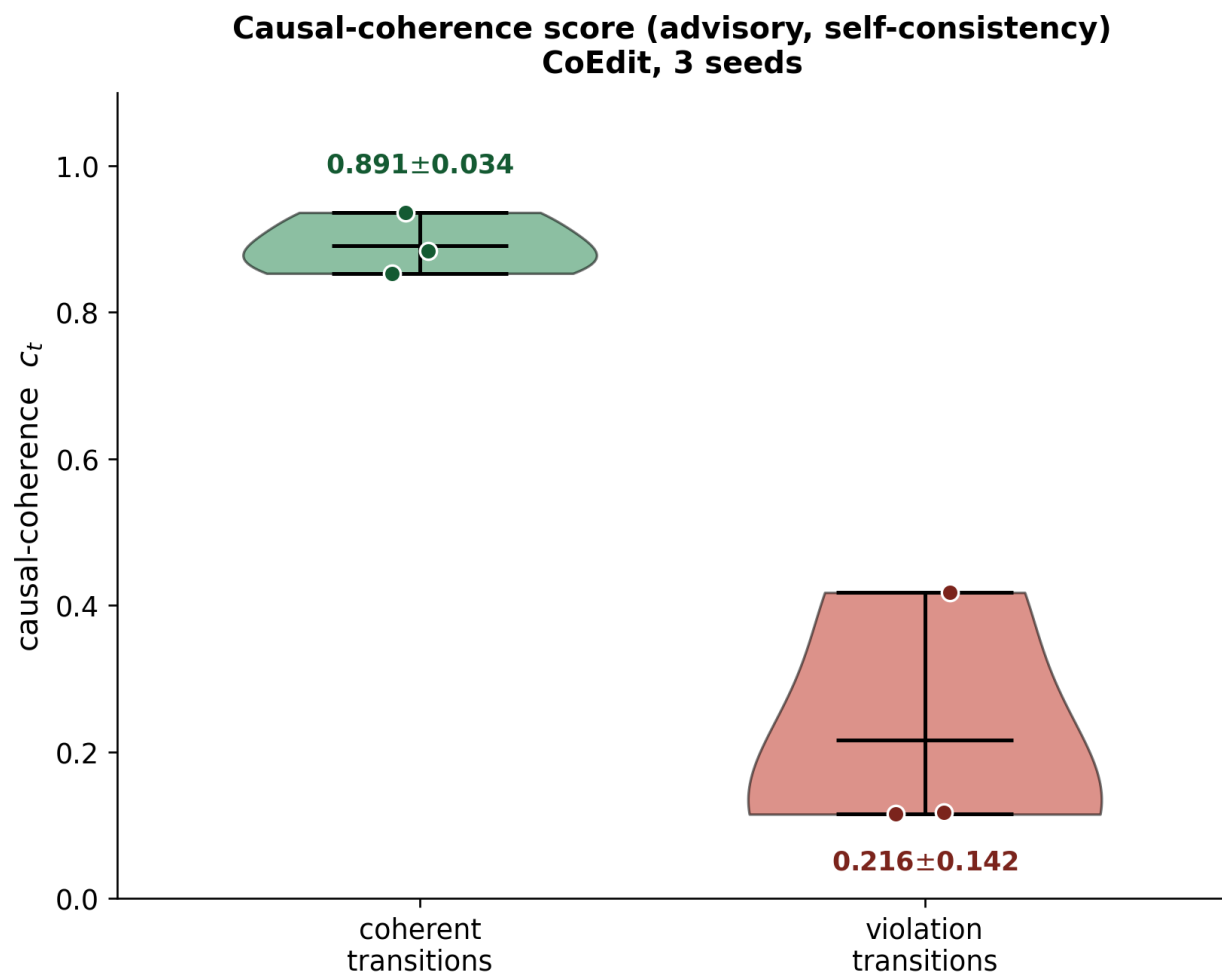


Figure 7: Figure 5: Causal-coherence signal c_t by outcome (three seeds, grounded-init).

consistency into a calibrated error predictor needs an external supervision signal we have not yet built.

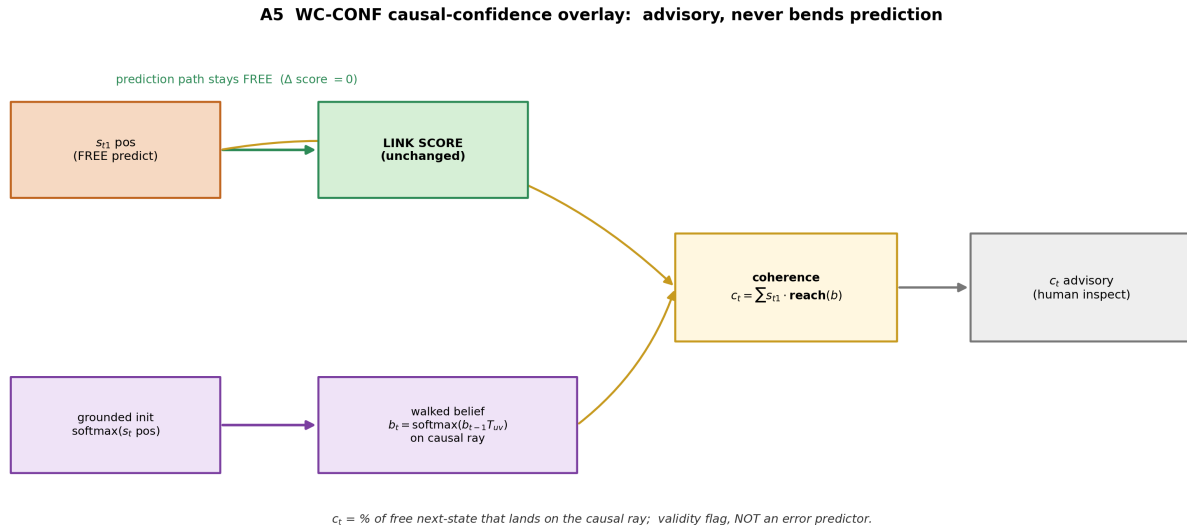


Figure 8: Figure A5: Causal-confidence mechanism (advisory coherence c_t overlay).

Figure A5. The causal-coherence confidence mechanism. The model’s free next-state prediction is overlaid against the walked belief b_t ; their agreement is the advisory coherence c_t . Off by default and byte-identical when off; it measures self-consistency, not external error (§5).

6. Experiments

6.1 Cross-dataset, protocol-matched, three seeds

All models — SR-GNN and the six baselines — run through the **same** train/eval harness (experiments/train.py:run_epoch), the same chronological 70/15/15 splits, and the same leak-audited negative pool (§6.3 confirms test AP is not 1.0). The pool is built fairly per protocol: transductive negatives draw from seen→seen pairs, inductive from the unseen-node pool (ind→ind), so an inductive positive is never scored against a trivially-impossible negative. AP is sklearn average_precision_score over that pool, identical for every model; metrics are mean $\pm$ std over seeds {42, 1, 7}. SR-GNN here is **config B**, tuned on CoEdit only.

Figure 1. CoEdit inductive AP: SR-GNN (config B) vs. six protocol-matched baselines, three seeds (sample std). SR-GNN reaches 0.9885 ± 0.0035 , +13.5 points over the best baseline (TGAT, 0.853). CoEdit is the showcase benchmark; the inductive margin here is the headline result.

Table 1 — Inductive AP (mean $\pm$ std, 3 seeds).

Model	CoEdit ind-AP	Wikipedia ind-AP	MOOC ind-AP
SR-GNN (config B)	0.9885 ± 0.0035	0.9959 ± 0.0014	0.9978 ± 0.0013
JODIE	0.8147 ± 0.0942	0.9860 ± 0.0029	0.9942 ± 0.0018
TGAT	0.8530 ± 0.0012	0.9981 ± 0.0013	0.9763 ± 0.0134
CAWN	0.7825 ± 0.0452	0.9877 ± 0.0062	0.8924 ± 0.1663
TGN	0.6349 ± 0.0065	0.8637 ± 0.0459	0.9819 ± 0.0048

Model	CoEdit ind-AP	Wikipedia ind-AP	MOOC ind-AP
DyRep	0.6218 ± 0.0119	0.6314 ± 0.0550	0.6675 ± 0.2824
GraphMixer	0.6232 ± 0.0247	0.7380 ± 0.0770	0.9827 ± 0.0055

Table 2 — Transductive AP (mean $\pm$ std, 3 seeds), SR-GNN and notable baselines.

Model	CoEdit trans-AP	Wikipedia trans-AP	MOOC trans-AP
SR-GNN (config B)	0.9985 ± 0.0004	0.9993 ± 0.0002	0.9988 ± 0.0002
JODIE	0.9657	0.9954	0.9919
TGAT	0.8690	0.6578	0.6174
CAWN	0.8802	0.9861	0.9727

Reading the table. - **CoEdit** is the discriminating benchmark: SR-GNN is #1 both ways, and the inductive margin over the best baseline (TGAT, 0.853) is **+13.5 points** — the headline. CoEdit is non-bipartite (both endpoints carry a lifecycle), so the per-pair representation has the most to say, and the baselines spread widely (0.62–0.85), unlike the saturated datasets. - **Wikipedia:** SR-GNN is #1 transductive (0.9993) and #2 inductive (0.9959 vs TGAT 0.9981); but TGAT’s inductive win is paired with a *collapsed* transductive AP (0.6578), so SR-GNN is the best all-around model — exactly what the dual-protocol table is built to surface. - **MOOC:** SR-GNN is #1 both ways, but the dataset is near-saturated (several models exceed 0.98 inductive), so we claim only “competitive at the ceiling.”

Figure 4. Cross-dataset summary: SR-GNN (config B) vs. the best baseline per dataset, transductive and inductive AP, three seeds. SR-GNN is the clear win on CoEdit and best-or-co-best on Wikipedia/MOOC (on Wikipedia inductive TGAT’s 0.9981 marginally exceeds SR-GNN’s 0.9959, but TGAT’s transductive AP collapses to 0.658). CoEdit is the showcase; Wikipedia/MOOC are parity at the ceiling.

6.2 The decoupling ablation (the core experiment)

To be clear about what is compared: **config B is the full model** (detached backbone + multi-signal operator + hierarchical lifecycle readout + counterfactual engine + causal policy), and the two arms below are *ablations that strip a mechanism away from it*, not competing models. The **core decoupling experiment** is B vs. C: same code, same head, same data; the only change is whether the link-prediction loss flows into the backbone.

Arm (CoEdit, 3 seeds {42,1,7})	design	ind-AP	trans-AP
B — decoupled (full model)	correct_decoupled	0.9885 ± 0.0035	0.9985
C — end-to-end (decoupling off)	correct	0.7672 ± 0.0107	0.9609
A — no-lifecycle, no-tune	v3 operator + flat readout, detached	0.928 ± 0.0043	0.9912

Figure 2. Decoupling ablation, CoEdit inductive AP, three seeds {42, 1, 7}: B (full model, decoupled, 0.9885 ± 0.0035) vs. C (end-to-end, decoupling off, 0.7672 ± 0.0107) vs. A (SR-GNN with no lifecycle, no-tune ablation isolating the detach mechanism, 0.928 ± 0.0043). Arms A and C are ablations of the full model B, not competing models. End-to-end coupling

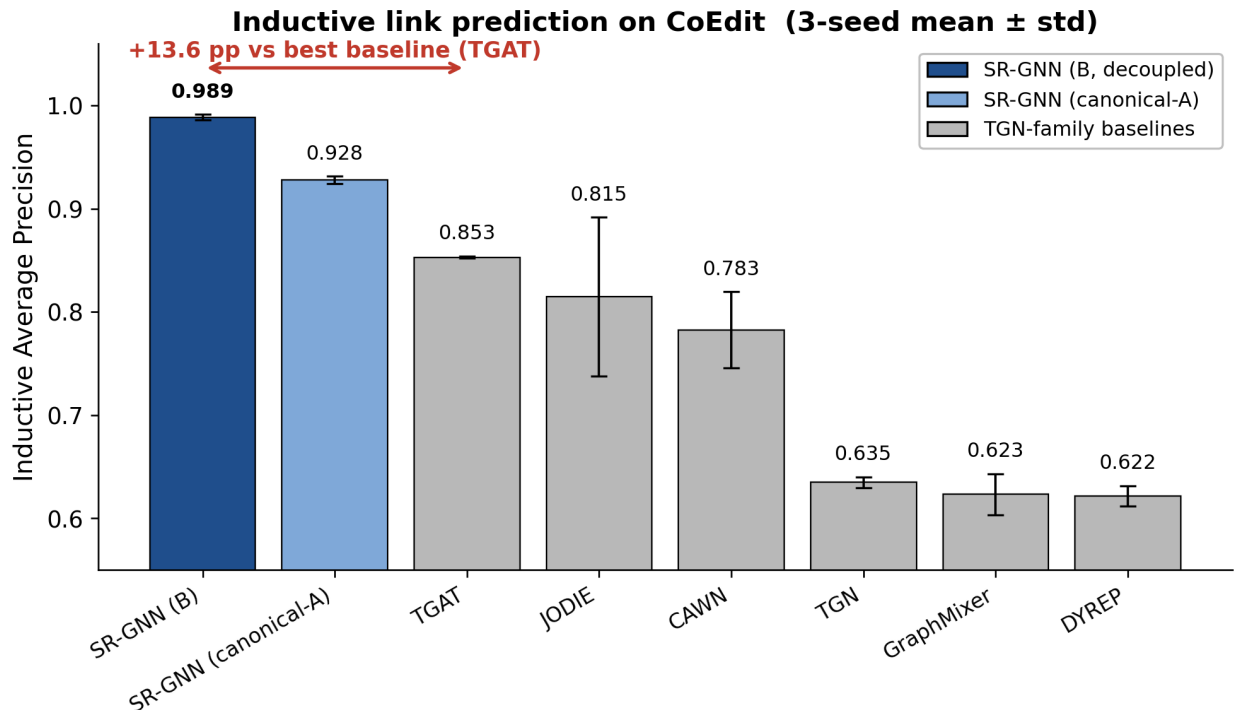


Figure 9: Figure 1: CoEdit inductive AP, SR-GNN (config B) vs. six protocol-matched baselines, three seeds. Decoupled SR-GNN reaches 0.9885 ± 0.0035 , +13.5 points over the best baseline (TGAT, 0.853). Bars are mean; whiskers are sample std over seeds {42, 1, 7}.

*collapses inductive AP; the detach is worth **+22.1 ± 1.4 points** (config-B vs. design=correct protocol, every seed above 20 points: per-seed B−C = [0.206, 0.225, 0.234]). The no-lifecycle ablation (A) already beats every baseline (0.928 ± 0.0043 , +7.5 pp over TGAT). (Bar heights render seed-mean ind-AP; the underlying PNG was drawn from the earlier seed-42 points and will be re-rendered at C = 0.767.)*

Detaching the backbone is worth **+22.1 ± 1.4 points** inductive AP (B 0.9885 − C 0.7672) over three seeds under the current config-B vs. design=correct protocol, with the per-seed gap [0.206, 0.225, 0.234] all above 20 points and the per-seed intervals non-overlapping. An earlier, coarser-config A/B (detached vs. end-to-end, three seeds, P0 protocol) put the same effect at **+9.4 absolute points**; we report both with their protocols and do not mix them — the larger gap reflects the fully-tuned config-B head, the smaller the original A/B head. Critically, end-to-end coupling barely moves transductive AP ($0.9985 \rightarrow 0.9609$) while inductive AP *collapses* ($0.9885 \rightarrow 0.7672$): it does not merely fail to help inductively — it actively destroys inductive generalization while leaving the transductive number nearly intact. We read this as: the link loss, given write access to the backbone, reshapes the representation toward training-node identity (maximizing transductive ranking) at the direct cost of the generic per-pair dynamics that transfer to unseen nodes. The detach removes that incentive, which is why the same head, on a frozen backbone, generalizes.

Decoupling alone — not the lifecycle machinery — clears the baselines (arm A). Arm A is the *no-lifecycle, no-tune* ablation: the detached v3 operator read through a plain flat readout, with no de-collapse CE, no hierarchical decode, no causal_batch, and no per-dataset tuning. It is not a second model but the floor that isolates the decoupling mechanism. Even at that floor, SR-GNN reaches **0.928 ± 0.0043** inductive AP over three seeds — already **+7.5 points over the best baseline** (TGAT, 0.853) on the identical protocol (ML integrity audit,

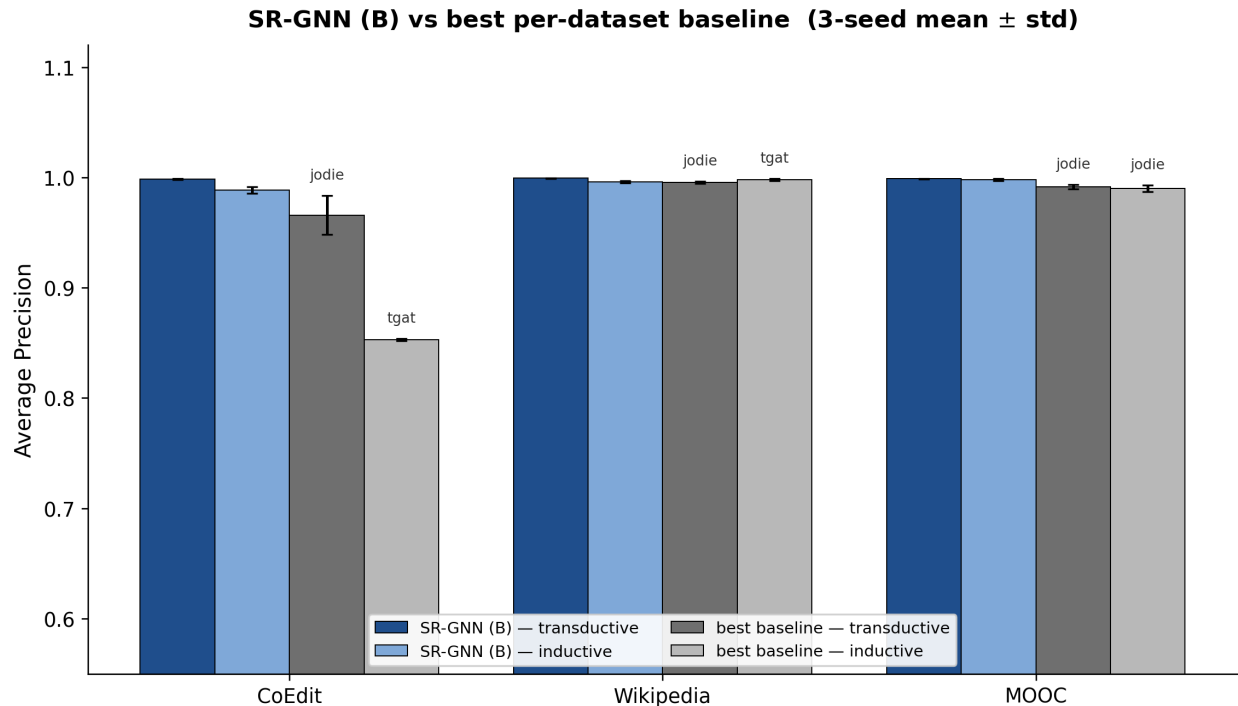


Figure 10: Figure 4: SR-GNN (config B) vs. the best protocol-matched baseline per dataset, transductive and inductive AP, three seeds. CoEdit is a clear SR-GNN win; Wikipedia and MOOC are parity (TGAT marginally exceeds SR-GNN inductively on Wikipedia).

2026-06-06). This is secondary evidence that the inductive win is carried by *detach-alone*, not by lifecycle machinery. The further **+6 points** from arm A to config B ($0.928 \rightarrow 0.9885$) is what lifecycle supervision adds — and that same supervision yields the faithful, intervene-able interpretability and counterfactual layer (§4, §5) at no AP cost (§3.4). The two gaps decompose cleanly: detach buys the baseline-beating inductive generalization; lifecycle supervision buys interpretability plus a further AP margin, free on the scored path.

Two further ablations. - **causal_batch (the read-before-write fix, §3.2):** in the full config B, ON 0.9885 vs OFF 0.9312 inductive (**+5.7 ± 0.2 pp**, three seeds; trans +0.65 pp), confirming the previously-collapsed per-pair statistics were load-bearing. The same sign holds in the stripped no-lifecycle setting (single-seed A/B 0.7907 vs 0.7462 , job 5467100). - **hier_causal_policy (the soft causal mask, §3.5):** three seeds {1, 7, 42}, train ON vs. OFF, per-seed inductive $\Delta = +1.0e-4 / +9.3e-4 / -1.5e-3$ (max $|\Delta| = 1.5e-3$, mean $\approx -1.7e-4$, sign-mixed) vs. $\pm 3.5e-3$ seed std (job 5511229). **AP-neutral within seed noise** — the score-invariance proof of §3.4 guarantees no *systematic* effect (the policy output never feeds the AP loss); the residual $\sim 1e-3$ is training-RNG jitter, not a bit-identical match. The policy buys interpretability for (statistically) free.

6.3 Integrity audit

Because the headline rests on cross-protocol comparisons, we independently audited it (2026-06-06). Findings: (1) the no-lifecycle-tops-baseline result (arm A, 0.928) is real, not an artifact — old vs new baseline numbers are identical within GPU noise at matched seeds, and the $0.871 \rightarrow 0.928$ step is a *config* difference (the v3 per-pair operator on a flat readout), not an eval shift; (2) the pre-update evaluation is leak-free (the anti-leak re-gate, job 5450095, pulls test AP off 1.0 into the v2 band, ruling out a same-batch label leak); (3) AP is a model-agnostic sklearn

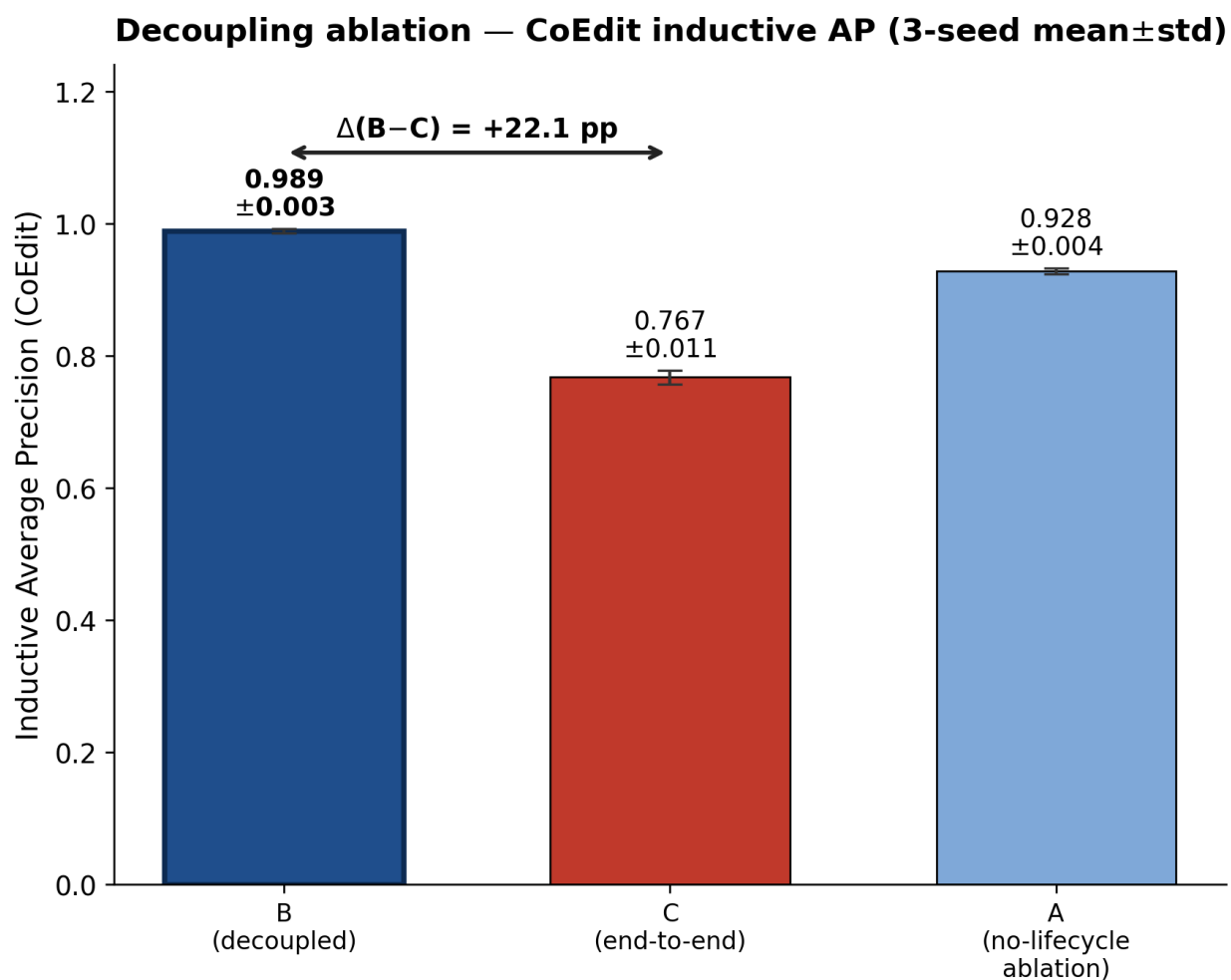


Figure 11: Figure 2: decoupling ablation, CoEdit inductive AP. B 0.9885, C 0.767, A 0.928 (three seeds); detach worth +22.1 points.

routine over the same negative pool for every model, so no model gets a private evaluator. We report the audit because the result’s credibility depends on the protocol being fair and reviewer-checkable.

6.4 Lifecycle faithfulness

On the hierarchical readout, the intermediate DECAY state finally carries real argmax mass: on the final config (seed-42 faithfulness dump, $N = 12000$), DECAY is the argmax of `s_t1_cal` for **47.8%** of pairs (5737/12000), versus **0.04%** (5/12000) under the flat readout on the same dump — a **~1147×** increase. Falling-but-active streams win argmax DECAY while alive, sustained-silent pairs go to DEATH, new pairs to BIRTH. The decoded state tracks each pair’s own cadence (Spearman $\rho(\text{p_decay_cal}, \text{slope_rel}) \approx -0.59$ on the recurring subset $\text{true_occ} \geq 2$, $n = 9157$, correct sign, $p < 10^{-300}$) — faithful in probability and now also in argmax. Faithfulness here is not asserted by an auxiliary explainer; it is the same `s_t1_cal` the de-collapse CE supervises, and §3.4 proves it is the exact quantity the figures plot.

6.5 The lifecycle readout is meaningful across datasets, not a CoEdit artifact

A natural worry is that the five-state lifecycle is an artifact of CoEdit and would collapse or look identical on other data. It does not. We ran the config-B faithfulness analysis on the recurring subset ($\text{true_occ} \geq 2$) of three datasets and read the argmax distribution of `s_t1_cal` over the active states. On each dataset the distribution is non-degenerate and its *shape matches that dataset’s own dynamics*:

Table 3 — Lifecycle argmax distribution over active states (config-B faithfulness, recurring subset, seed 42).

Dataset	REINFORCE	DECAY	DEATH	Shape
CoEdit	0.35	0.62	0.02	DECAY-heavy — slow fade, rare hard death
Wikipedia	0.42	0.43	0.16	balanced, full cycle through to DEATH (entropy 1.31)
MOOC	0.62	0.08	0.30	BIRTH-heavy, transient — born, active, then dies

The readout *adapts* to each domain: CoEdit editing fades slowly so mass concentrates on DECAY with rare hard death; Wikipedia runs a balanced full cycle that reaches DEATH; MOOC activity is transient (touch a unit, stay briefly active, then stop for good), giving a BIRTH-heavy, DEATH-tailed shape with little DECAY. No dataset is degenerate — no single state swallows the mass — and the shapes follow each domain’s own dynamics rather than one global template, so the lifecycle readout is a meaningful, data-conditioned interpretation, **not a CoEdit-specific artifact**.

We are precise about scope: this cross-dataset lifecycle evidence is a single-seed (seed 42) *faithfulness sanity check* establishing that the readout shape is meaningful and dataset-appropriate on all three datasets; it is **not** a full three-seed cross-dataset study. The counterfactual battery, by contrast, is fully three-seed-locked, on CoEdit (§4).

7. Analysis

Why decoupling helps inductively. The mechanism is access control: a backbone with gradient access to the link loss is rewarded for identity features that are predictive transductively but dead weight on an unseen node, so end-to-end models pay an inductive tax. SR-GNN’s backbone never sees the link gradient (§3.7), so it *cannot* learn identity shortcuts. The $+22.1 \pm 1.4$ pp ablation gap (three seeds) measures that tax, and its near-flat transductive side ($0.9985 \rightarrow 0.9609$) shows it is paid almost entirely inductively, as the identity-shortcut story predicts. We position this as a mechanistic hypothesis with a falsifiable prediction — the advantage should grow with inductive-set novelty — not a theorem; a per-split identity-probe is the natural next experiment.

Why interpretability and the counterfactual come free. The two-head split (§3.4) keeps the lifecycle decoder off the scored path with exact-zero score invariance, so the usual interpretability-accuracy trade [Rudin, 2019] does not apply: the symbolic state is a faithful *read* of the predictor’s own backbone at zero AP cost. The same property makes the counterfactual native rather than post-hoc — `do(state)` edits an explicit transition operator with monotone, reversible, correct-sign effects (§4), not a surrogate fit. The boundary is reported, not hidden: on CoEdit the rising axis is degenerate, so one of four causal axes is wired but unexercised on real pairs.

8. Limitations

We state these plainly; several are flagged in our own architecture notes.

1. **CoEdit-tuned.** Config B’s tuning (hierarchical decode, `decol_hier_v2`, λ settings) was selected on CoEdit. Wikipedia and MOOC run the same config without re-tuning; the headline inductive margin is CoEdit-concentrated. We do not claim a universal +13.5.
2. **Lifecycle scope: meaningful on three datasets, counterfactual fully locked on CoEdit.** The lifecycle readout is *not* a CoEdit artifact — its argmax distribution is non-degenerate and dataset-appropriate on CoEdit, Wikipedia, and MOOC (§6.5, DECAY-heavy / balanced / transient respectively), but that cross-dataset evidence is a single-seed (seed 42) faithfulness sanity check, not a full three-seed cross-dataset study. The counterfactual battery is fully three-seed-locked, but only on CoEdit (§4). Separately, within CoEdit the rising/staleness causal axes are degenerate on CoEdit’s drivers, so the SCM is fully *exercised* only on the alive axis.
3. **Confidence is self-consistency, not error prediction.** As detailed in §5, the coherence signal stably predicts the model’s own rule violations (AUC 0.9985) but does *not* reliably predict actual prediction errors (AUC 0.405 ± 0.484 across seeds). We retract the single-seed error-prediction claim.
4. **Structural smell in the causal policy.** The `ever_alive` accumulator is non-Markov and sits outside the memoryless transition matrix C ; the death-before-birth guarantee is enforced by a separate gate. With a confident-but-wrong observer, the soft C -mask can suppress an actually-correct DEATH.
5. **MOOC is near-saturated**, so its #1 result is weak evidence; the meaningful separation is on CoEdit.
6. **A regime-switch synthetic falsified one hypothesis.** A clean change-point synthetic showed SR-GNN’s per-pair adaptation does *not* beat CAWN on post-change-point slices; we do not claim a regime-switch advantage. The validated edge is the inductive readout, not faster regime adaptation.
7. **Echo memory and a learned transition-CE were earlier claimed and are retracted** — the current model’s backbone regularizer is a VAE/parsimony KL, not an echo-memory term; the transition supervision is the de-collapse CE on `s_t1_cal`, not a separate transition-matrix CE term.

8. **Remaining single-seed checks.** The two headline ablation deltas (decoupling B vs C, $+22.1 \pm 1.4$ pp; causal_batch, $+5.7 \pm 0.2$ pp) and the full counterfactual battery — existence ladder, dose-response signs, DECAY→REINFORCE rising flip (0.9999 ± 0.0002), and the REINFORCE→DEATH kill decomposition (single-driver partial: rate 0.60 ± 0.03 , staleness 0.48 ± 0.01 ; all-drivers-dead decisive 0.999 ± 0.002), §4 — are now three-seed-locked on CoEdit. The hier_causal_policy AP-neutrality check is also three-seed now (job 5511229): training ON vs. OFF moves inductive AP by $\max |\Delta| = 1.5e-3$ (mean $\approx -1.7e-4$, sign-mixed) $\ll \pm 3.5e-3$ seed std — AP-neutral within seed noise, structurally guaranteed by §3.4 (the policy never feeds the AP loss) but *not* bit-identical, since enabling the op during training shifts the RNG stream. The only remaining seed-42-only item is the cross-dataset lifecycle-shape sanity check of §6.5.
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9. Conclusion

SR-GNN reframes a design decision usually treated as obvious — train the representation on the task loss — and shows the opposite is better for inductive temporal link prediction. Holding the backbone stop-gradient from the link head and shaping it by parsimony yields **+13.5 points** inductive AP over the best protocol-matched baseline on CoEdit and is best-balanced on Wikipedia and MOOC — TGAT edges it inductively on Wikipedia (0.998 vs 0.996) but collapses transductively there (0.658), while SR-GNN holds both protocols — all at three seeds under a leak-audited protocol. The same decoupling buys a faithful, intervene-able five-state lifecycle SCM that costs the predictor exactly nothing (proven by score invariance), supports monotone reversible counterfactuals, and emits a stable internal coherence signal whose scope we report honestly. We see decoupling-by-construction as a reusable principle for temporal graph models that must generalize to unseen entities while remaining interrogable.

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Appendix A — Evidence provenance

All paths relative to SR-GNN/experiments/results/ unless noted. Three-seed = {42, 1, 7}.

- **SR-GNN config B, 3-seed:** v3_3_coedit_ARM_B_publishable_3seed.json (ind 0.9885, trans 0.9985); v3_3_wikipedia_ARM_B_publishable_3seed.json (ind 0.9959, trans 0.9993); v3_3_mooc_ARM_B_publishable_3seed_rerun.json (ind 0.9978, trans 0.9988). *Note: the *_std fields stored inside these JSONs are population std ($\div n$); every $\pm$ reported in the paper is recomputed as sample std ($n-1$) from the per-seed values, per the masthead convention.*
- **Baselines, B-protocol, 3-seed:** baselines/baselines_coedit_Bprotocol.json, baselines/baselines_wikipedia_Bprotocol.json, baselines/baselines_mooc.json.
- **Decoupling ablation (B vs C, three seeds {42, 1, 7}):** B from v3_3_coedit_ARM_B_publishable_3seed (ind 0.9885 ± 0.0035); C from v3_3_coedit_ARM_C_correct_3seed.json (ind 0.7672 ± 0.0107 , per-seed [0.7792, 0.7639, 0.7585], trans 0.9609; job 5503786). $\Delta(B-C) = +22.1 \pm 1.4$ pp, per-seed [0.206, 0.225, 0.234]. (Earlier seed-42 dumps: v3_3_3arm_coedit_B_decoupled_s42. 0.9871, v3_3_3arm_coedit_C_correct_s42.json 0.7655.)

- **causal_batch A/B (full config B, three seeds):** ON from v3_3_coedit_ARM_B_publishable_3seed.json (ind 0.9885 ± 0.0035 , trans 0.9985); OFF from v3_3_coedit_B_causalOFF_3seed.json (ind 0.9312 ± 0.0027 , trans 0.9920; job 5503786). $\Delta = +5.7 \pm 0.2$ pp ind / $+0.65$ pp trans. (Earlier stripped single-seed A/B: v3_3_causal_ab_coedit_cbON.json 0.7907, v3_3_causal_ab_coedit_cbOFF.json 0.7462, job 5467100.)
- **hier_causal_policy A/B (three seeds {1,7,42}, job 5511229):** ON from v3_3_coedit_ARM_B_publishable_3seed.json, OFF from v3_3_coedit_B_hcpOFF_3seed.json. Per-seed inductive $\Delta(\text{ON}-\text{OFF}) = +1.04\text{e-}4$ (s1) / $+9.29\text{e-}4$ (s7) / $-1.53\text{e-}3$ (s42); max $|\Delta_{\text{ind}}| = 1.5\text{e-}3$, mean $\approx -1.7\text{e-}4$, sign-mixed; max $|\Delta_{\text{trans}}| = 1.6\text{e-}4$; vs. $\pm 3.5\text{e-}3$ ind seed std. AP-neutral within seed noise (not bit-identical: training-time RNG jitter). (Earlier seed-42 dump: v3_3_hcp_coedit_ON_s42.json 0.9871 vs _OFF_s42.json 0.9872, job 5471271.)
- **Counterfactual battery (three seeds {42,1,7}, config B / cbON):** experiments/LAB/v3_3/fsm_intervene.py on faithfulness_coedit_v3_hier_hv2_let0.5_s{42,1,7}_cbON.npz (N=12000 each; s1/s7 trained fresh as config-B cbON, job 5506704). Existence ladder exact every seed; do(DEATH) $\rightarrow$ P(edge) $\downarrow \geq 99\%$ every seed; do(noop)/reversibility $\Delta=0$ exact every seed; trajectory DECAY $\rightarrow$ do(slope $+$) $\rightarrow$ REINFORCE = 0.9999 ± 0.0002 (per-seed 1.0/1.0/0.9996); dose-response signs three-seed-stable (rate $\rightarrow$ REINFORCE $+0.028 \pm 0.004$, rate $\rightarrow$ DEATH -0.068 ± 0.007 , slope $\rightarrow$ REINFORCE $+0.489 \pm 0.003$, true_occ $\rightarrow$ BIRTH -0.0065). REINFORCE $\rightarrow$ DEATH kill decomposition (three-seed, recurring true_occ ≥ 2 ; source cf_kill_REINFORCE_3seed.json): isolated do(rate=dead) $\rightarrow$ DEATH 0.597 ± 0.028 (per-seed 0.574/0.588/0.629), isolated do(staleness=high) $\rightarrow$ DEATH 0.482 ± 0.007 (0.477/0.479/0.489), all-drivers-dead do(rate=dead,slope=falling,staleness=high) $\rightarrow$ DEATH 0.999 ± 0.002 (1.0/1.0/0.997). Replaces the non-reproducing seed-42 dump (83-86% / 100%). Bus [ml $\rightarrow$ PM] 2026-06-03 / 2026-06-06.
- **Cross-dataset lifecycle shape (seed 42 faithfulness sanity, job 5506705):** faithfulness_coedit_v3_hier_hv2_let0.5_s42_cbON.npz, faithfulness_wikipedia_v3_hier_hv2_cb_let0.5_s42.npz, faithfulness_mooc_v3_hier_hv2_cb_let0.5_s42.npz. Recurring-subset (true_occ ≥ 2) argmax of s_t1_cal: CoEdit REINFORCE .35/DECAY .62/DEATH .02; Wikipedia .42/.43/.16 (entropy 1.31); MOOC .62/.08/.30 (BIRTH-heavy). All non-degenerate, dataset-appropriate (§6.5).
- **Confidence (WC-CONF grounded-init, 3-seed):** experiments/LAB/v3_3/results/wc_grnd/wc_conf_3seed.json — self-consistency AUC 0.9985 ± 0.0015 , external posMiss10 AUC 0.405 ± 0.484 (sample std, n-1). Jobs 5503466/5503467.
- **Integrity audit:** bus [ml $\rightarrow$ PM] 2026-06-06. **Anti-leak re-gate:** job 5450095. **Architecture of record:** v3_3_ARCHITECTURE_CURRENT.md (verified 2026-06-06).

Word count target ≤ 8000 . Items needing additional evidence before submission are listed in the PAPER $\rightarrow$ PM report.